

Data-Efficient Real-Time Control of an Artificial-Muscle-Driven Continuum Robot with Physics-Informed Koopman Operator

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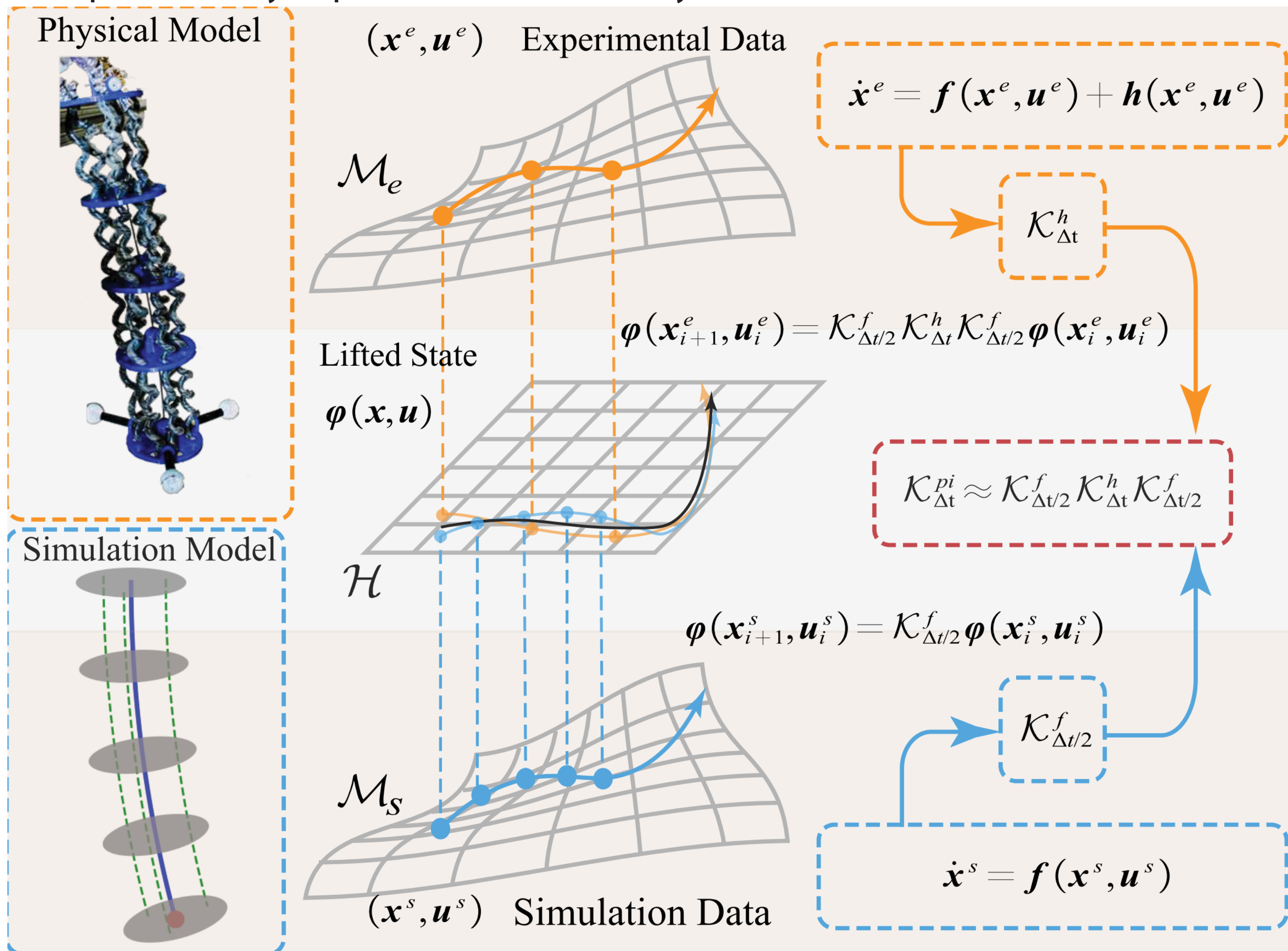
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Citation: J. Wang, E. Ristich, E. Weissman, Y. Ren and J. Sun, in IEEE Robotics and Automation Letters, vol. 11, no. 4, pp. 5080-5087, April 2026

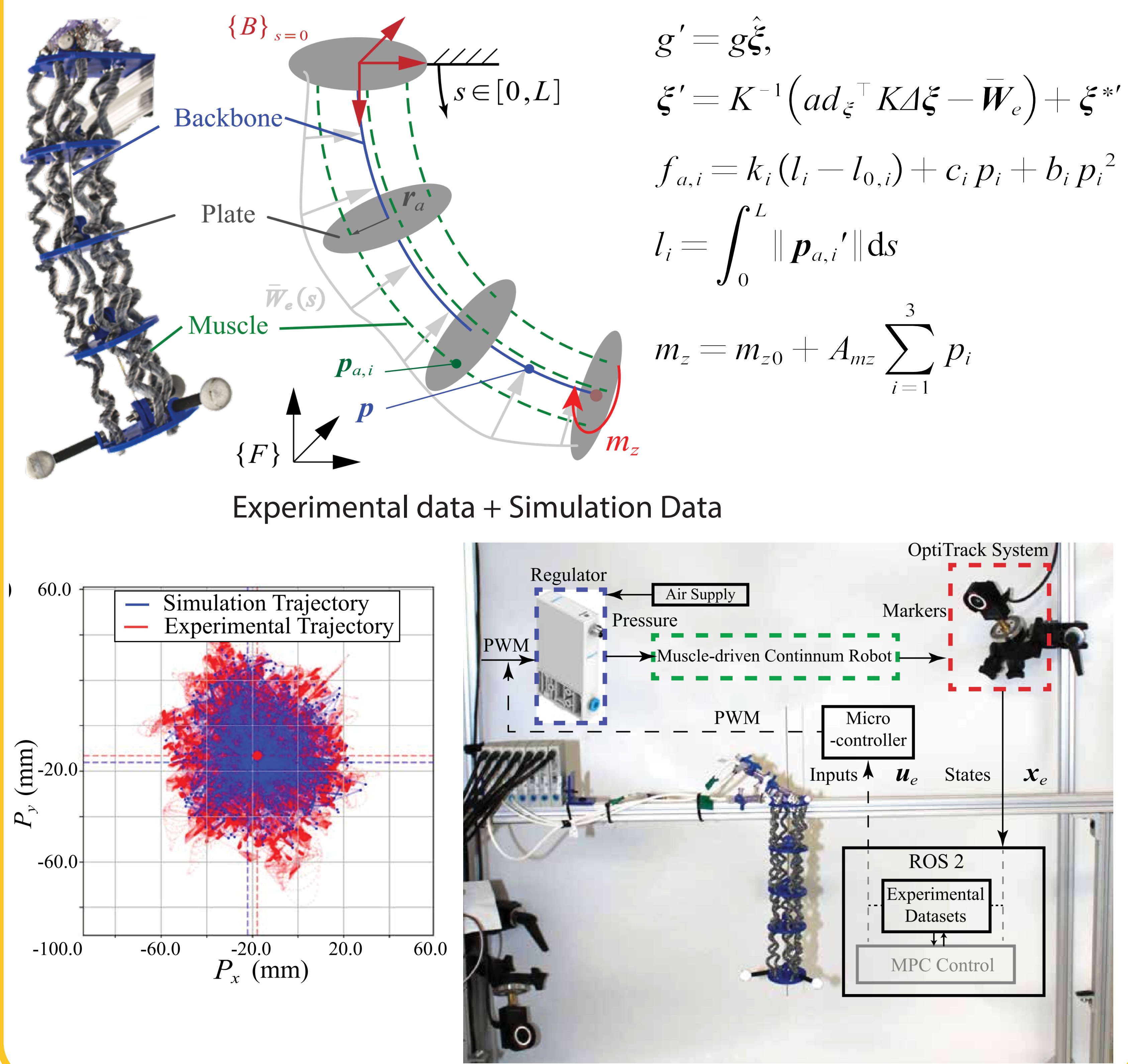
Introduction

Trade-off between physics-based and data-driven modeling:
computationally expensive VS data costly

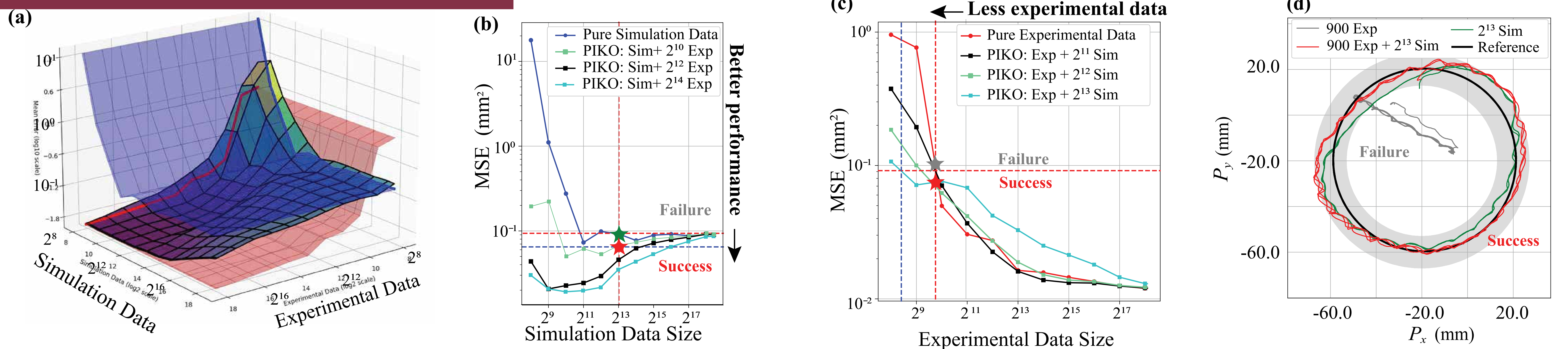


In this work, we implement the physics-informed Koopman operator (PIKO) framework on an FR-PTCA-driven continuum robot, **integrating prior physical knowledge** from a static rod-muscle coupled model to **reduce experimental data requirements**.

Artificial-muscle-driven continuum robot

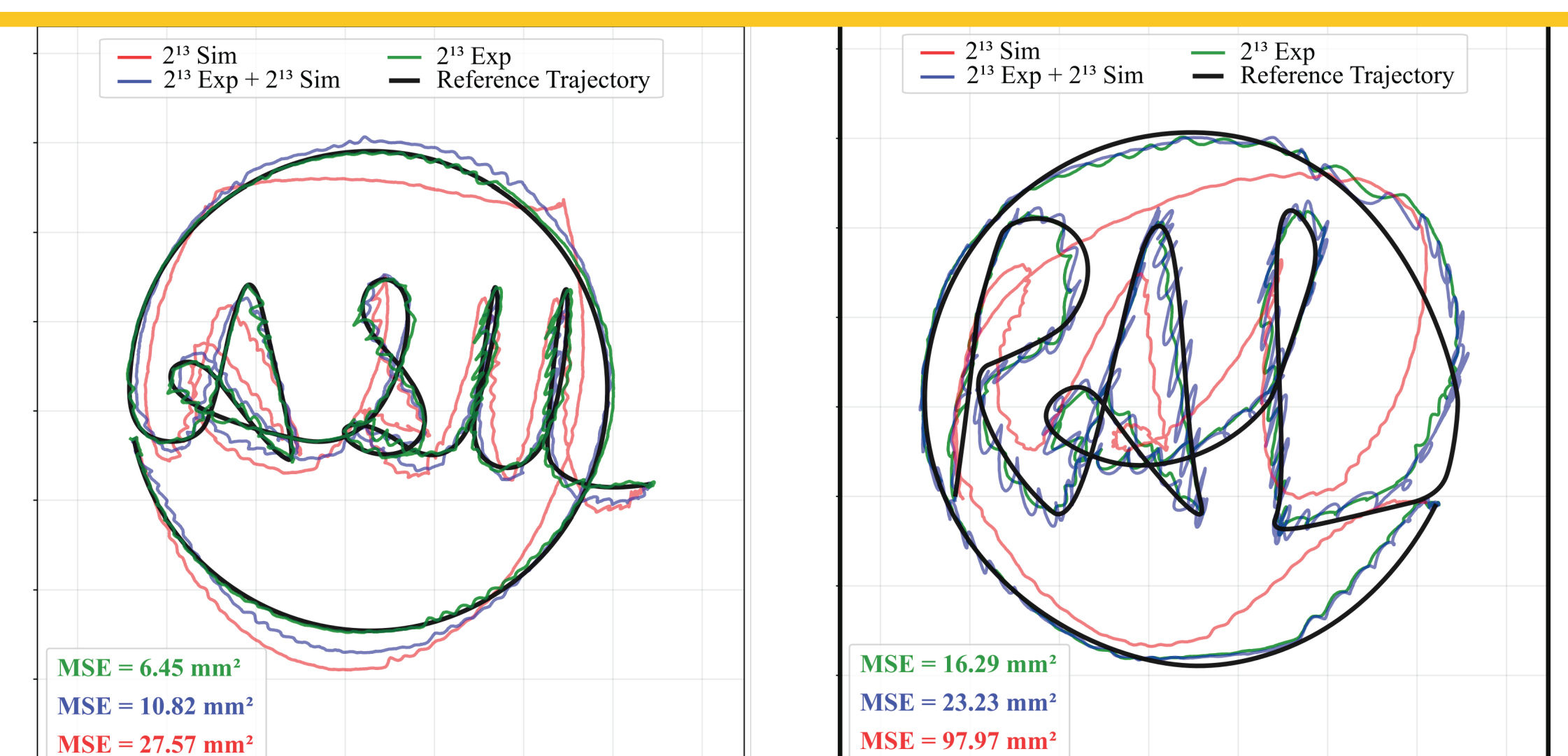


PIKO training & validation



Discussion & Conclusion

- Applying the PIKO framework to an artificial-muscle-driven continuum robot, where **the available physics-based model is limited to a static model**;
- Demonstrating **successful MPC-based closed-loop control with limited experimental data**, which fails without physics-informed prior knowledge
- Systematically comparing** models trained with varying amounts of simulation and experimental data to highlight the advantages of PIKO in low-data regimes.



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